PROGRAM 01:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main()
{
    pid_t pid;
    pid = fork();   // create child process

    if (pid < 0) {
        printf("Fork failed\n");
    }
    else if (pid == 0) {
        printf("Child Process\n");
        printf("Child PID = %d\n", getpid());
        printf("Parent PID = %d\n", getppid());
    }
    else
{
            wait(NULL);   // wait for child to finish
        printf("Parent Process\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }
   return 0;
```

}

OUTPUT:

Child Process

Child PID = 12068

Parent PID = 12067

Parent Process

Parent PID = 12067

Child PID = 12068

PROGRAM 02:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>
int main()
{
    int fd;
    char write_buf[] = "Hello, I/O System Call in C\n";
    char read_buf[50];
    fd = open("sample.txt", O_CREAT | O_RDWR, 0644);

    if (fd < 0) {
        printf("File open failed\n");
        return 1;
    }
    write(fd, write_buf, sizeof(write_buf));

    lseek(fd, 0, SEEK_SET);
    read(fd, read_buf, sizeof(read_buf));
    printf("Data read from file:\n%s", read_buf);
    close(fd);
    return 0;
}
```

OUTPUT:

Data read from file:

Hello, I/O System Call in C

PROGRAM 03:

```
#include <stdio.h>
#include <unistd.h>
#include <string.h>
int main()
{
    int fd[2];
    pid_t pid;
    char write_msg[] = "Hello from Parent Process";
    char read_msg[50];
    if (pipe(fd) == -1) {
        printf("Pipe creation failed\n");
        return 1;
    }
    pid = fork();
    if (pid < 0) {
        printf("Fork failed\n");
        return 1;
    }
    if (pid > 0) {

        close(fd[0]);  // Close read end
        write(fd[1], write_msg, strlen(write_msg) + 1);
        close(fd[1]);  // Close write end
```

```
    }
    else
   {

        close(fd[1]);  // Close write end
        read(fd[0], read_msg, sizeof(read_msg));
        printf("Child received message: %s\n", read_msg);
        close(fd[0]);  // Close read end
    }

    return 0;
}
```

OUTPUT:

Child received message: Hello from Parent Process

PROGRAM 04:

```
#include<stdio.h>
void main()
{
int bt[20], wt[20], tat[20], i, n;
float wtavg, tatavg;
printf("\nEnter the number of processes -- ");
scanf("%d", &n);
for(i=0;i<n;i++)
{
printf("\nEnter Burst Time for Process %d -- ", i);
scanf("%d", &bt[i]);
}
wt[0] = wtavg = 0;
tat[0] = tatavg = bt[0];
for(i=1;i<n;i++)
{
wt[i] = wt[i-1] +bt[i-1];
tat[i] = tat[i-1] +bt[i];
wtavg = wtavg + wt[i];
tatavg = tatavg + tat[i];
}
printf("\t PROCESS \tBURST TIME \t WAITING TIME\t TURNAROUND
TIME\n");
for(i=0;i<n;i++)
```

```
printf("\n\t P%d \t\t %d \t\t %d \t\t %d", i, bt[i], wt[i], tat[i]);
printf("\nAverage Waiting Time -- %f", wtavg/n);
printf("\nAverage Turnaround Time -- %f", tatavg/n);

}
```

OUTPUT:

Enter the number of processes -- 4

Enter Burst Time for Process 0 -- 2

Enter Burst Time for Process 1 -- 5

Enter Burst Time for Process 2 -- 7

Enter Burst Time for Process 3 -- 8

PROCESS	BURST TIME	WAITING TIME	TURNAROUND TIME
P0	2	0	2
P1	5	2	7
P2	7	7	14
P3	8	14	22

Average Waiting Time -- 5.750000

Average Turnaround Time -- 11.250000

PROGRAM 05:

```
#include<stdio.h>
void main()
{
int p[20], bt[20], wt[20], tat[20], i, k, n, temp; float wtavg,
tatavg;
printf("\nEnter the number of processes -- ");
scanf("%d", &n);
for(i=0;i<n;i++)
{
p[i]=i;
printf("Enter Burst Time for Process %d -- ", i);
scanf("%d", &bt[i]);
}
for(i=0;i<n;i++)
for(k=i+1;k<n;k++)
if(bt[i]>bt[k])
{
temp=bt[i];
bt[i]=bt[k];
bt[k]=temp;
temp=p[i];
p[i]=p[k];
p[k]=temp;
}
```

```
wt[0] = wtavg = 0;

tat[0] = tatavg = bt[0]; for(i=1;i<n;i++)

{

wt[i] = wt[i-1] +bt[i-1];

tat[i] = tat[i-1] +bt[i];

wtavg = wtavg + wt[i];

tatavg = tatavg + tat[i];

}

printf("\n\t PROCESS \tBURST TIME \t WAITING TIME\t
TURNAROUND TIME\n");

for(i=0;i<n;i++)

printf("\n\t P%d \t\t %d \t\t %d \t\t %d", p[i], bt[i], wt[i], tat[i]);

printf("\nAverage Waiting Time -- %f", wtavg/n);

printf("\nAverage Turnaround Time -- %f", tatavg/n);

}
```

OUTPUT:

Enter the number of processes -- 5

Enter Burst Time for Process 0 -- 2

Enter Burst Time for Process 1 -- 5

Enter Burst Time for Process 2 -- 7

Enter Burst Time for Process 3 -- 9

Enter Burst Time for Process 4 -- 3

PROCESS	BURST TIME	WAITING TIME	TURNAROUND TIME
P0	2	0	2
P4	3	2	5
P1	5	5	10
P2	7	10	17
P3	9	17	26

Average Waiting Time -- 6.800000

Average Turnaround Time -- 12.000000

PROGRAM 06:

```
#include<stdio.h>
void main()
{
int p[20],bt[20],pri[20], wt[20],tat[20],i, k, n, temp; float wtavg,tatavg;
printf("Enter the number of processes --- ");
scanf("%d",&n);
for(i=0;i<n;i++){
p[i] = i;
printf("Enter the Burst Time & Priority of Process %d --- ",i);
scanf("%d %d",&bt[i], &pri[i]);
}
for(i=0;i<n;i++)
for(k=i+1;k<n;k++)
if(pri[i] > pri[k]){
temp=p[i];
p[i]=p[k];
p[k]=temp;
temp=bt[i];
bt[i]=bt[k];
bt[k]=temp;
temp=pri[i];
pri[i]=pri[k];
pri[k]=temp;
}
```

```
wtavg = wt[0] = 0;

tatavg = tat[0] = bt[0];

for(i=1;i<n;i++)

{

wt[i] = wt[i-1] + bt[i-1];

tat[i] = tat[i-1] + bt[i];

wtavg = wtavg + wt[i];

tatavg = tatavg + tat[i];

}

printf("\nPROCESS\t\tPRIORITY\tBURST TIME\tWAITING
TIME\tTURNAROUND TIME");

for(i=0;i<n;i++)

printf("\n%d \t\t %d \t\t %d \t\t %d \t\t %d ",p[i],pri[i],bt[i],wt[i],tat[i]);

printf("\nAverage Waiting Time is --- %f",wtavg/n);

printf("\nAverage Turnaround Time is --- %f",tatavg/n);

}
```

OUTPUT:

Enter the number of processes --- 5

Enter the Burst Time & Priority of Process 0 --- 3

5

Enter the Burst Time & Priority of Process 1 --- 7

8

Enter the Burst Time & Priority of Process 2 --- 3

4

Enter the Burst Time & Priority of Process 3 --- 5

6

Enter the Burst Time & Priority of Process 4 --- 7

5

PROCESS	PRIORITY	BURST TIME	WAITING TIME	TURNAROUND TIME
2	4	3	0	3
0	5	3	3	6
4	5	7	6	13
3	6	5	13	18
1	8	7	18	25

Average Waiting Time is --- 8.000000

Average Turnaround Time is --- 13.00000

PROGRAM 07:

```
#include<stdio.h>
void main()
{
int i,j,n,bu[10],wa[10],tat[10],t,ct[10],max;
float awt=0,att=0,temp=0;

printf("Enter the no of processes -- ");
scanf("%d",&n);
for(i=0;i<n;i++)
{
printf("\nEnter Burst Time for process %d -- ", i+1);
scanf("%d",&bu[i]);
ct[i]=bu[i];
}
printf("\nEnter the size of time slice -- ");
scanf("%d",&t);
max=bu[0];
for(i=1;i<n;i++)
if(max<bu[i])
max=bu[i];
for(j=0;j<(max/t)+1;j++)
for(i=0;i<n;i++)
if(bu[i]!=0)
if(bu[i]<=t) {
```

```
tat[i]=temp+bu[i];
temp=temp+bu[i];
bu[i]=0;
}
else {
bu[i]=bu[i]-t;
temp=temp+t;
}
for(i=0;i<n;i++){
wa[i]=tat[i]-
ct[i]; att+=tat[i];
awt+=wa[i];}
printf("\nThe Average Turnaround time is -- %f",att/n);
printf("\nThe Average Waiting time is -- %f ",awt/n);
printf("\n\tPROCESS\t BURST TIME \t WAITING TIME\tTURNAROUND
TIME\n");
for(i=0;i<n;i++)
printf("\t%d \t %d \t\t %d \t\t %d \n",i+1,ct[i],wa[i],tat[i]);
}
```

OUTPUT:

Enter the no of processes -- 5

Enter Burst Time for process 1 -- 3

Enter Burst Time for process 2 -- 2

Enter Burst Time for process 3 -- 5

Enter Burst Time for process 4 -- 8

Enter Burst Time for process 5 -- 6

Enter the size of time slice -- 3

The Average Turnaround time is -- 14.000000

The Average Waiting time is -- 9.200000

PROCESS	BURST TIME	WAITING TIME	TURNAROUND TIME
1	3	0	3
2	2	3	5
3	5	11	16
4	8	16	24
5	6	16	22

PROGRAM 08 :

```
#include<stdio.h>
int main()
{
int i,j,n,a[50],frame[10],no,k,avail,count=0;
        printf("\n ENTER THE NUMBER OF PAGES:\n");
scanf("%d",&n);
        printf("\n ENTER THE PAGE NUMBER :\n");
        for(i=1;i<=n;i++)
        scanf("%d",&a[i]);
        printf("\n ENTER THE NUMBER OF FRAMES :");
        scanf("%d",&no);
for(i=0;i<no;i++)
        frame[i]= -1;
                j=0;
                printf("\tref string\t page frames\n");
for(i=1;i<=n;i++)
                {
                        printf("%d\t\t",a[i]);
                        avail=0;
                        for(k=0;k<no;k++)
if(frame[k]==a[i])
                                avail=1;
                        if (avail==0)
                        {
```

```
                                        frame[j]=a[i];
                                        j=(j+1)%no;
                                        count++;
                                        for(k=0;k<no;k++)
                                        printf("%d\t",frame[k]);
}
                                printf("\n");
}
                        printf("Page Fault Is %d",count);
                        return 0;
}
```

OUTPUT:

ENTER THE NUMBER OF PAGES:

4

ENTER THE PAGE NUMBER :

4

5

7

2

ENTER THE NUMBER OF FRAMES :3

ref string	page frames		
4	4	-1	-1
5	4	5	-1
7	4	5	7
2	2	5	7

Page Fault Is 4

PROGRAM 09 :

```
#include<stdio.h>
void main()
{
int q[20],p[50],c=0,c1,d,f,i,j,k=0,n,r,t,b[20],c2[20];
printf("Enter no of pages:");
scanf("%d",&n);
printf("Enter the reference string:");
for(i=0;i<n;i++)
        scanf("%d",&p[i]);
printf("Enter no of frames:");
scanf("%d",&f);
q[k]=p[k];
printf("\n\t%d\n",q[k]);
c++;
k++;
for(i=1;i<n;i++)
        {
                c1=0;
                for(j=0;j<f;j++)
                {
                        if(p[i]!=q[j])
                        c1++;
                }
                if(c1==f)
```

```
{
        c++;
        if(k<f)
        {
                q[k]=p[i];
                k++;
                for(j=0;j<k;j++)
                printf("\t%d",q[j]);
                printf("\n");
        }
        else
        {
                for(r=0;r<f;r++)
                {
                        c2[r]=0;
                        for(j=i-1;j<n;j--)
                        {
                        if(q[r]!=p[j])
                        c2[r]++;
                        else
                        break;
                }
        }
        for(r=0;r<f;r++)
         b[r]=c2[r];
        for(r=0;r<f;r++)
```

```
                {
                    for(j=r;j<f;j++)
                    {
                        if(b[r]<b[j])
                        {
                            t=b[r];
                            b[r]=b[j];
                            b[j]=t;
                        }
                    }
                }
                for(r=0;r<f;r++)
                {
                    if(c2[r]==b[0])
                    q[r]=p[i];
                    printf("\t%d",q[r]);
                }
                printf("\n");
            }
        }
    }
    printf("\nThe no of page faults is %d",c);
    }
```

OUTPUT:

```
Enter number of frames: 3
Enter number of pages: 7
Enter page reference string:
1 3 0 3 5 6 3
Page    Frames
1       1 - -
3       1 3 -
0       1 3 0
3       1 3 0
5       5 3 0
6       5 6 0
3       5 6 3

Total Page Faults = 6
```

PROGRAM 10 :

```
#include<stdio.h>
int main()
{
int n,pg[30],fr[10];
int count[10],i,j,k,fault,f,flag,temp,current,c,dist,max,m,cnt,p,x;
fault=0;
dist=0;
k=0;
printf("Enter the total no pages:\t");
scanf("%d",&n);
printf("Enter the sequence:");
for(i=0;i<n;i++)
scanf("%d",&pg[i]);
printf("\nEnter frame size:");
scanf("%d",&f);

for(i=0;i<f;i++)
{
count[i]=0;
fr[i]=-1;
}
for(i=0;i<n;i++)
{
flag=0;
```

```
temp=pg[i];
for(j=0;j<f;j++)
{
if(temp==fr[j])
{
flag=1;
break;
}
}
if((flag==0)&&(k<f))
{
fault++;
fr[k]=temp;
k++;
}
else if((flag==0)&&(k==f))
{
fault++;
for(cnt=0;cnt<f;cnt++)
{
current=fr[cnt];
for(c=i;c<n;c++)
{
if(current!=pg[c])
count[cnt]++;
else
```

```
break;
}
}
max=0;
for(m=0;m<f;m++)
{
if(count[m]>max)
{
max=count[m];
p=m;
}
}
fr[p]=temp;
}
printf("\npage  %d  frame\t",pg[i]);
for(x=0;x<f;x++)
{
printf("%d\t",fr[x]);
}
}
printf("\nTotal number of faults=%d",fault);
return 0;
}
```

OUTPUT:

Enter the total no pages: 5

Enter the sequence:3

5

7

8

9

Enter frame size:3

page	3	frame	3	-1	-1
page	5	frame	3	5	-1
page	7	frame	3	5	7
page	8	frame	8	5	7
page	9	frame	9	5	7

Total number of faults=5